

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: **Nexora**

Project / Product Name: **Smart Guided Troubleshooting Engine**

Organization / Institution (if any): **Thapar Institute Of Engineering and Technology**

Submission Date: **30-09-2026**

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project?

Yes

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: Used Claude and ChatGPT to brainstorm the 5-phase RAG architecture and fallback logic.

Code generation or assistance: Used Open-AI Codex and GitHub Copilot to write the hybrid retriever and semantic cache logic.

UI / UX design: Used Google Gemini to generate custom CSS for a Samsung One UI-themed Stream-lit dashboard.

Content creation: Used Notebook Lm and Gemini to structure the `README .MD` and generate presentation slides.

Data analysis: N/A

Testing / debugging: Used Gemini to debug strict Pydantic validation errors and resolve API latency bottlenecks.

Other: Used ChatGPT to generate Mermaid code for system architecture flowcharts.

4. Feature Origin Classification:

1. Feature Name: Hybrid Retrieval Engine & Semantic Cache

2. Self-Generated/AI-Generated/Both: Both

3. Description:

AI Tools Used: Open-AI Codex, GitHub Copilot.

- **Prompt Used:** *"Write a Python RAG retriever combining BM25 and Sentence-Transformers. Also, implement a two-stage semantic cache for fast hits."*
- **Output Summary:** Generated Python classes for dense/sparse searching and in-memory caching.
- **Modification:** Manually integrated the code into our Fast-API routes and tweaked the similarity thresholds to ensure <15ms latency.

1. Feature Name: Samsung One UI Frontend Dashboard

2. Self-Generated/AI-Generated/Both: Both

3. Description:

- **AI Tools Used:** Google Gemini.
- **Prompt Used:** *"Create custom CSS for a Stream-lit app that perfectly mimics the Samsung One UI design language, including rounded cards and Samsung Blue."*
- **Output Summary:** Provided a complete `streamlit_app.py` file with embedded HTML/CSS for a polished frontend.
- **Modification:** Manually connected the frontend inputs to our backend API endpoints and programmed the live latency metrics display.

1. Feature Name: Safety Validators & Deterministic Post-Processor

2. Self-Generated/AI-Generated/Both: Both

3. Description:

- **AI Tools Used:** ChatGPT, Claude.
- **Prompt Used:** *"Write Pytest and Pydantic validators that catch LLM hallucinations, strip fake URLs, and enforce 5-7 word action descriptions."*
- **Output Summary:** Generated regex patterns and strict sorting algorithms (Auto → Manual → Critical).
- **Modification:** Hand-tuned the error handling to gracefully return a `no_match` fallback as required by the hackathon specifications.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: KRISHNA KAPOOR

Role: TEAM LEADER

Signature: 

Date: 30-09-2026